

Quantum Gates

Purpose. This material develops the intuition and mathematical language needed to understand common single-qubit and two-qubit gates, and connects each idea to Qiskit circuits and visualization.

1 The one idea to remember

A quantum gate is an operation that changes a quantum state. A single qubit can be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

The numbers α and β are amplitudes. Measuring in the computational basis gives

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

A gate changes amplitudes, including their phase. It is therefore not simply a classical operation that directly changes probabilities.

Computational basis

State	Vector	Bloch-sphere location
$ 0\rangle$	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	North pole (+Z)
$ 1\rangle$	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	South pole (−Z)

2 Bloch-sphere intuition

The Bloch sphere gives a geometric picture of a pure single-qubit state. The north pole represents $|0\rangle$ and the south pole represents $|1\rangle$. The equator contains equal-magnitude superpositions with different relative phases.

- X , Y and Z are π rotations about the corresponding Bloch-sphere axes.
- $R_x(\theta)$, $R_y(\theta)$ and $R_z(\theta)$ are controllable rotations.
- Phase gates such as S and T change relative phase and correspond to Z -axis rotations.

3 Pauli-X: the quantum NOT

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Therefore

$$X |0\rangle = |1\rangle, \quad X |1\rangle = |0\rangle.$$

Memory rule: X flips the qubit. On the Bloch sphere, X is a π rotation around X .

4 Pauli-Y

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}.$$

Thus

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle.$$

The Y gate is a π rotation around the Y axis. Unlike a classical NOT, it also introduces phase factors.

5 Pauli-Z

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Hence

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

The Z gate does not swap the computational basis states. It changes the phase of the $|1\rangle$ component. Its effect becomes especially important when later gates cause interference.

6 Hadamard gate

The Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Its most important actions are

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle,$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

Starting from $|0\rangle$, the Hadamard gate creates a coherent equal superposition. Measurement in the computational basis then gives 0 and 1 with probability $1/2$ each.

Do not think of H as simply “randomly choosing 0 or 1.” Think of it as creating a coherent superposition. The measurement happens later.

An important identity is

$$H^2 = I.$$

Therefore applying H twice returns the original state.

7 Phase gates: S and T

The S gate is

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix},$$

and the T gate is

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}.$$

They add phase to the $|1\rangle$ component without changing its magnitude. Their effect is particularly useful when the qubit is already in a superposition.

8 Rotation gates

Rotation gates allow us to choose the rotation angle θ :

$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix},$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix},$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}.$$

Gate	Intuition	Axis
$R_x(\theta)$	rotate by θ	X
$R_y(\theta)$	rotate by θ	Y
$R_z(\theta)$	rotate by θ	Z

In Qiskit, the angle is specified in radians.

At $\theta = \pi$,

$$R_x(\pi) = -iX, \quad R_y(\pi) = -iY, \quad R_z(\pi) = -iZ.$$

The factor $-i$ is a global phase, so these operations have the same observable single-qubit measurement probabilities as their corresponding Pauli gates.

9 Two-qubit gates: CNOT / CX

A CNOT has a **control** qubit and a **target** qubit. The target is flipped only when the control is $|1\rangle$.

Control	Target	Output
0	0	00
0	1	01
1	0	11
1	1	10

Memory rule: control = “should I act?” target = “what gets acted on?”

In Qiskit, the operation is written as

```
qc.cx(control, target).
```

CNOT is also a fundamental entangling gate.

10 H + CNOT and entanglement

Starting from $|00\rangle$, apply H to qubit 0:

$$|00\rangle \longrightarrow \frac{|00\rangle + |10\rangle}{\sqrt{2}}.$$

Then apply CNOT with qubit 0 as control and qubit 1 as target:

$$\frac{|00\rangle + |10\rangle}{\sqrt{2}} \rightarrow \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

This is a Bell state. In the ideal computational-basis measurement, the outcomes are correlated: 00 or 11.

11 Statevector, Bloch sphere and measurement

These three viewpoints answer different questions:

- **Statevector:** what are the amplitudes and phases?
- **Bloch sphere:** what is the geometric orientation of a single qubit?
- **Measurement:** what classical outcomes do we actually observe?

For the Qiskit exercises, the recommended workflow is:

1. Prepare the initial state.
2. Apply one or more gates.
3. Inspect the statevector.
4. Use `plot_bloch_multivector` for single-qubit geometry.
5. Use `visualize_transition` to see how a circuit transforms the state.
6. Measure when you want classical outcomes.

12 Gate reference table

Gate	Meaning	Geometric / operational view
X	Flip $ 0\rangle \leftrightarrow 1\rangle$	π rotation about X
Y	Flip with phase factors	π rotation about Y
Z	Add a minus sign to $ 1\rangle$	π rotation about Z
H	Create/remove equal superposition	Maps computational basis to X basis
S	Add $\pi/2$ phase to $ 1\rangle$	Z -axis phase rotation
T	Add $\pi/4$ phase to $ 1\rangle$	Z -axis phase rotation
$R_x(\theta)$	Rotation by θ	X axis
$R_y(\theta)$	Rotation by θ	Y axis
$R_z(\theta)$	Rotation by θ	Z axis
CNOT/CX	Flip target if control is 1	Two-qubit conditional gate

13 Practice questions

1. Start with $|0\rangle$ and apply X . What is the final state?
2. Start with $|0\rangle$ and apply H . What are the amplitudes and measurement probabilities?
3. Start with $|0\rangle$ and apply H twice. Explain the result.
4. Start with $|0\rangle$ and apply H , then Z , then H . Predict the final state.

5. Start with $|0\rangle$ and apply $R_x(\pi)$. Compare it with applying X .
6. Start with $|00\rangle$ and apply H to qubit 0 followed by $\text{CNOT}(0,1)$. What measurement outcomes should appear ideally?

14 A useful mental model

X/Y/Z: think π rotations.

H: think superposition and a change of basis.

S/T: think phase changes around Z .

$R_x/R_y/R_z$: think controllable rotations by θ .

CNOT: think “if control is 1, flip target.”

The companion Qiskit notebook turns each of these ideas into small executable experiments using `statevectors`, `plot_bloch_multivector`, and `visualize_transition`.